# Pilot experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

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## Before you start

Recall Predefining an analysis plan. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

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## Your mission

> Start with the target: Explain the mechanism and terminology of pilot experiments. Try the recall before opening notes.

### Your learning target

I can explain the **mechanism** and terminology of **pilot** experiments.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Predefining an analysis plan. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Research ethics and consent. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Ablation studies. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of **pilot** experiments

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Retrieve what you already know

Recall Predefining an analysis plan. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

### This week's end goal

Explain **pilot** experiments using the records and boundaries of this fictional investigation.

## Notice the evidence

> Investigate the supplied Pilot experiments case. Separate observations, given facts and assumptions before choosing an explanation.

A **pilot** checks **feasibility**, measurement and failure handling before the primary study. **Pilot** results used for revision are development **evidence**.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

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STUDENT LESSON · 3 OF 18

## See how it works

> Follow the mechanism for Pilot experiments. Connect each labelled input, deciding step and result.

Run a small fixture set to test data capture, timing boundaries, rubric and analysis code. Fix defects and freeze the revised plan before new primary cases.

**Pilot experiments — evidence trace**

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Pilot five cases reveals wrong units in duration logs; team corrects units and reports the pilot as final evidence. | Run a small fixture set to test data capture, timing boundaries, rubric and analysis code. Fix defects and freeze the revised plan before new primary cases. | Preserve pilot defect and correction. Pilot informed the method; use newly reserved primary cases for a final comparison. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

### Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

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STUDENT LESSON · 4 OF 18

## Compare the cases

> Compare the supplied cases for Pilot experiments. Identify the detail that changes the conclusion.

### Compare a familiar case with a changed case

**Pilot** five cases reveals wrong units in duration logs; team corrects units and reports the **pilot** as final **evidence**. Show the intermediate steps and give the conclusion with its scope.

**Keep the procedure fixed. Explain what changed in the inputs and result.**

### Change one thing in the pictured case

#### Change one thing

Change: Add the corrected **pilot** answers into final evaluation.

#### Predicted result

**Prediction**: Declare them previously exposed rather than untouched final **evidence**.

#### Why the result follows

Reason: Rubric development already used those answers.

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STUDENT LESSON · 5 OF 18

## Words that unlock the idea

Use the vocabulary of Pilot experiments to explain a concrete decision, rather than only memorising definitions.

| Word | Meaning |
| --- | --- |
| Pilot | A preliminary study of the procedure |
| Feasibility | Whether the planned method can be carried out |
| Measurement defect | A failure in collecting the intended outcome |

**Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

## Follow a worked example

> Follow the worked trace for Pilot experiments, predict a step, then compare with its explanation.

**Pilot** five cases reveals wrong units in duration logs; team corrects units and reports the **pilot** as final **evidence**.

Preserve **pilot** defect and correction. **Pilot** informed the method; use newly reserved primary cases for a final comparison.

### Worked explanation · Pilot experiments

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

> The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Pilot experiments

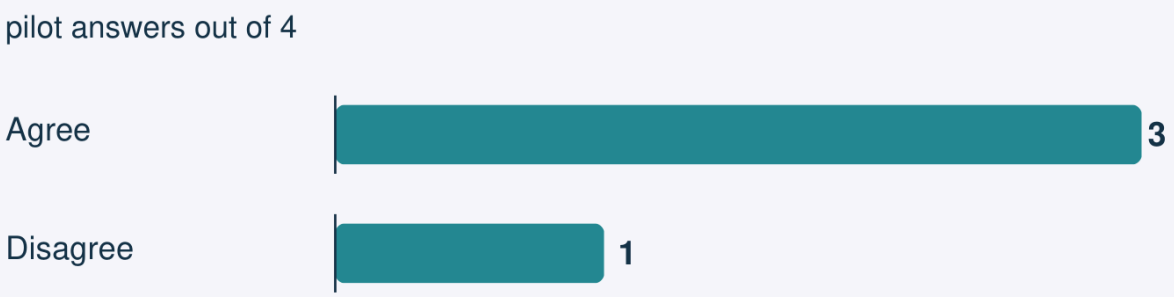
A **pilot** tests the procedure. A **pilot** checks whether the team can score answers consistently before running the main study.

### The facts we will use

> **Supplied facts**
>
> Two reviewers score four practice answers. They agree on three and disagree on one; these answers are excluded from final evaluation.

AI ACADEMY / AI-7/22

## Reviewer agreement on four pilot answers




Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/22

# Pilot experiments

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Try the workflow

Both reviewers apply the draft rubric to four practice answers.



## 2. Measure agreement

Agreement is 3/4=75% on this pilot.



## 3. Repair the rubric

Discuss the disagreement and clarify the scoring rule before the main run.

**CHANGE ONE THING**

Add the corrected pilot answers into final evaluation.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Try the workflow | Both reviewers apply the draft rubric to four practice answers. |
| 2. Measure agreement | Agreement is 3/4=75% on this pilot. |
| 3. Repair the rubric | Discuss the disagreement and clarify the scoring rule before the main run. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

#### Why the result follows

A **pilot** can reveal missing instructions, practical delays or ambiguous scoring. It is preparation, not an untouched final estimate after the team uses it to revise methods.

### What this example does not establish

#### Keep the stated limit

Four **pilot** answers do not establish long-run reviewer reliability or a stable performance effect.

### Keep the stated limit

Read the labelled picture: Calculate reviewer agreement on the four **pilot** answers. What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?

### Keep the stated limit

Change one thing: Add the corrected **pilot** answers into final evaluation. Predict the result before checking the explanation. What remains the same?

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STUDENT LESSON · 7 OF 18

## Find and repair a mistake

> Find the unsupported step in the Pilot experiments claim and explain a repair using the supplied evidence.

A **pilot** may assess **feasibility** but is not automatically powered or independent **evidence** for the final hypothesis.

### Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

### Locate the earliest unsupported step

A report omits this qualification: A **pilot** may assess **feasibility** but is not automatically powered or independent **evidence** for the final hypothesis.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**State one repair and the checks you would rerun.**

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STUDENT LESSON · 8 OF 18

## Practise together

> Practise Pilot experiments with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

**Explain your prediction to a partner before comparing results.**

### Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Pilot** experiments". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
|---|---|---|
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for **Pilot** experiments. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

> Calculate reviewer agreement on the four **pilot** answers. What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?

**My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

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STUDENT LESSON · 9 OF 18

## Run the investigation

> Investigate Pilot experiments. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

### Predict, investigate and preserve the evidence

**Pilot** five cases reveals wrong units in duration logs; team corrects units and reports the **pilot** as final **evidence**. Show the intermediate steps and give the conclusion with its scope.

| Record | Your evidence |
| --- | --- |
| Given input and fixed rule | |
| Prediction before checking | |
| Actual result, or “not run” | |

| Record | Your evidence |
| --- | --- |
| First mismatch and one explanation | |
| Boundary or missing input and expected response | |

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

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STUDENT LESSON · 10 OF 18

## Build your understanding

> Move from recall to tracing, diagnosis and transfer in Pilot experiments. Explain why each step is justified.

**Explain → apply → challenge**

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: A **pilot** may assess **feasibility** but is not automatically powered or independent **evidence** for the final hypothesis.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

## Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled **pilot** experiments **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **pilot** experiments.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

| Project evidence | Record |
| --- | --- |
| Component and version | |
| Inputs permitted at prediction time | |
| Trace and comparison with earlier work | |
| Failure, limit and human review | |

### Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

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STUDENT LESSON · 12 OF 18

## Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My limited claim and a useful next check**

STUDENT LESSON · 13 OF 18

# Study a complete case

> Revisit the complete Pilot experiments case for supported practice. Repeated examples do not become fresh assessments.

### Complete evidence record

**Pilot experiments — evidence trace**

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Pilot five cases reveals wrong units in duration logs; team corrects units and reports the pilot as final evidence. | Run a small fixture set to test data capture, timing boundaries, rubric and analysis code. Fix defects and freeze the revised plan before new primary cases. | Preserve pilot defect and correction. Pilot informed the method; use newly reserved primary cases for a final comparison. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

**My complete trace of the worked case**

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STUDENT LESSON · 14 OF 18

## Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

### Independent case: Apply a new case independently

**Pilot** six cases exposes ambiguous labels in two. What should happen before the primary study?

**My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Attempt this without hints

> **Pilot** six cases exposes ambiguous labels in two. What should happen before the primary study?

**My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

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STUDENT LESSON · 15 OF 18

## Your lesson route

> Use this map to navigate Pilot experiments, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A **pilot** tests the procedure. A **pilot** checks whether the team can score answers consistently before running the main study.

| By the end, I can | How I will show it |
|---|---|
| Explain the mechanism and terminology of pilot experiments | Explain the deciding step, show a trace or test, and state one limit. |
| Apply the stated procedure to a new case with a traceable result | Explain the deciding step, show a trace or test, and state one limit. |
| Identify an assumption or limitation and design a revealing follow-up | Explain the deciding step, show a trace or test, and state one limit. |

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

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STUDENT LESSON · 16 OF 18

## Finish and return

> Explain Pilot experiments without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **pilot** experiments

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

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STUDENT LESSON · 17 OF 18

## Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A **pilot** tests the procedure. A **pilot** checks whether the team can score answers consistently before running the main study.

**Practice 1: explain the deciding evidence**

Which labelled step matches this detail? Discuss the disagreement and clarify the scoring rule before the main run.

1. Try the workflow

2. Repair the rubric

3. Measure agreement

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Repair the rubric Discuss the disagreement and clarify the scoring rule before the main run.

A **pilot** can reveal missing instructions, practical delays or ambiguous scoring. It is preparation, not an untouched final estimate after the team uses it to revise methods.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: Add the corrected **pilot** answers into final evaluation. Which conclusion is supported by the supplied facts?

1. Declare them previously exposed rather than untouched final **evidence**.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Declare them previously exposed rather than untouched final **evidence**. Rubric development already used those answers.

**Repair a missing building block**

Review Predefining an analysis plan (ai-7/21). Goal: Explain the **mechanism** and terminology of predefining an analysis plan. Short practice: Apply the recorded exclusion rule.

Can two readable model failures be removed, and how must a necessary later plan change be disclosed?. Return to this week and explain what you changed.

Review Research ethics and consent (ai-7/19). Goal: Explain the **mechanism** and terminology of research ethics and consent. Short practice: Identify which uses the drawing consent permits and excludes.

Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?. Return to this week and explain what you changed.

Review Ablation studies (ai-7/16). Goal: Explain the **mechanism** and terminology of ablation studies. Short practice: Calculate the filter ablation's two rates and count change.

Which single component is altered, and why does also changing the model prevent isolating the filter?. Return to this week and explain what you changed.

**Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

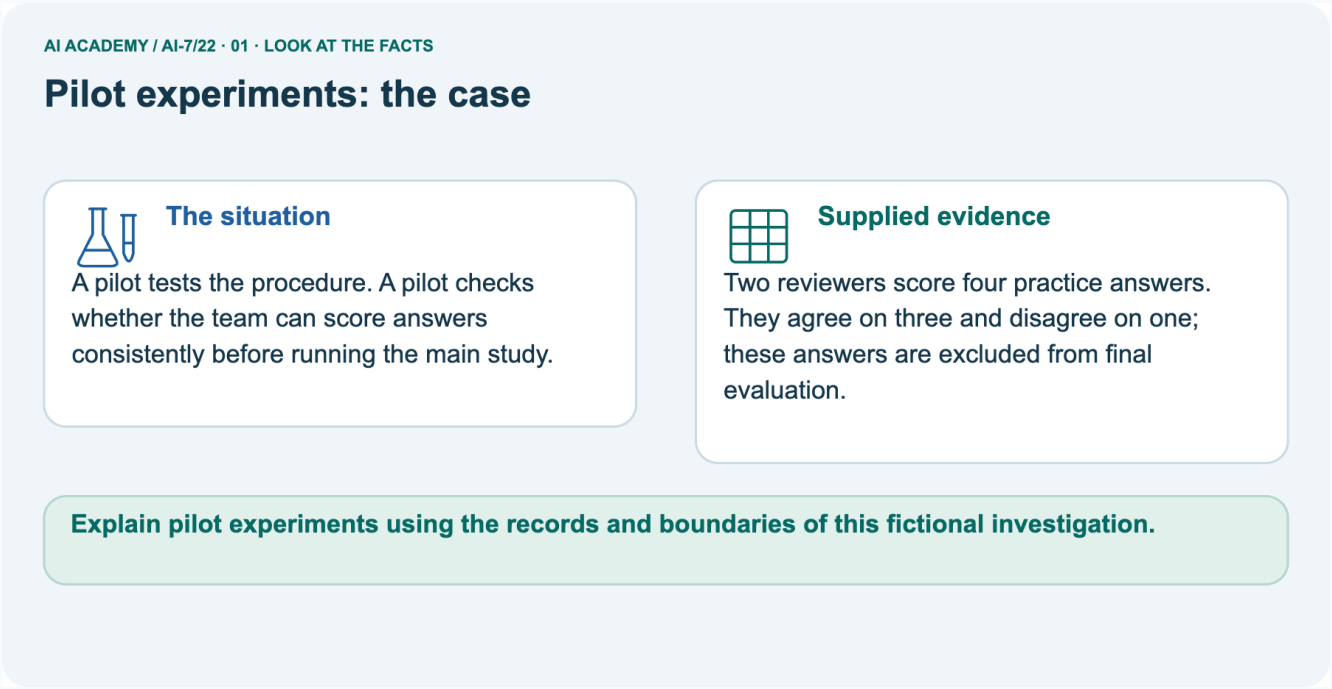
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STUDENT LESSON · 18 OF 18

## Visual concept journey

> Explain each Pilot experiments graphic using its labels, deciding evidence and relevant limitation.

## 01. Pilot experiments: the case



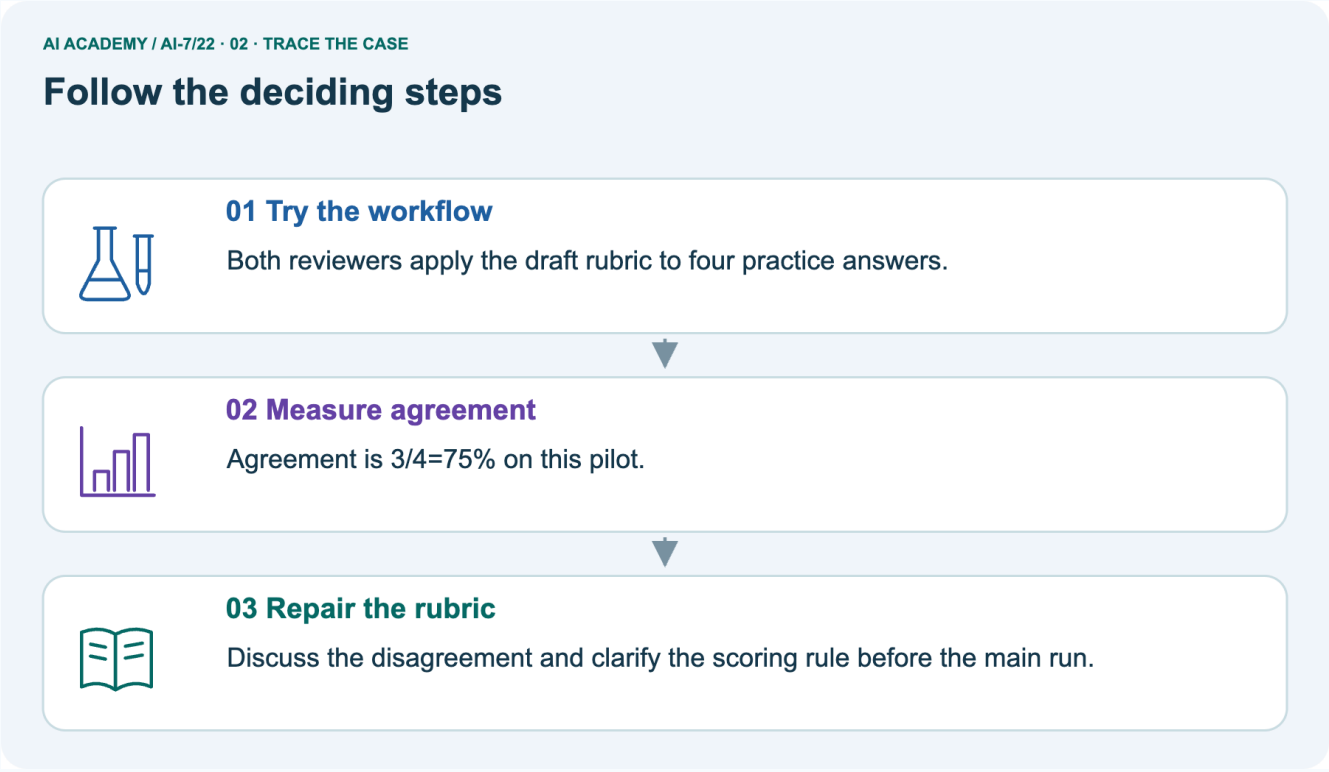

Follow the labelled picture with the matching explanation. Open the vector diagram

A **pilot** tests the procedure. A **pilot** checks whether the team can score answers consistently before running the main study. Supplied facts: Two reviewers score four practice answers.

They agree on three and disagree on one; these answers are excluded from final evaluation.

> Point to the given fact that matters most. Explain why it matters.

## 02. Follow the deciding steps

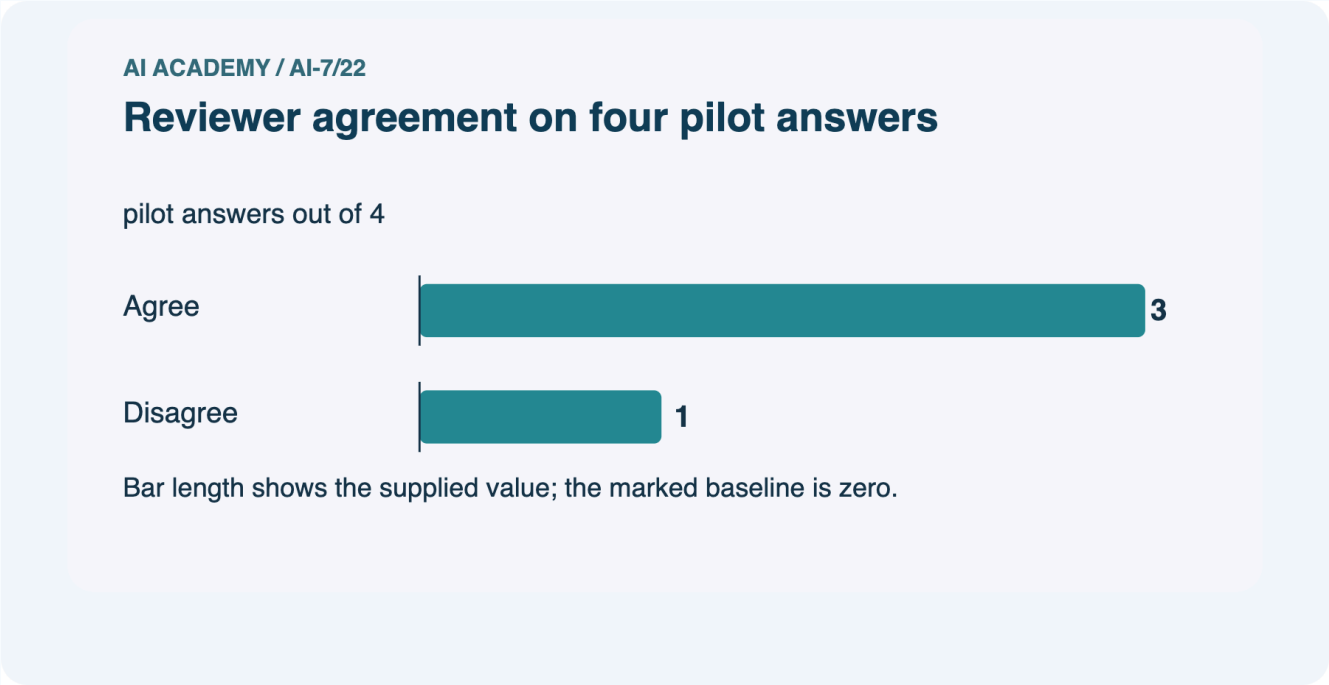



Follow the labelled picture with the matching explanation. Open the vector diagram

Try the workflow: Both reviewers apply the draft rubric to four practice answers. → Measure agreement: Agreement is 3/4=75% on this **pilot**. → Repair the rubric: Discuss the disagreement and clarify the scoring rule before the main run.

> Point to each numbered stage and explain which recorded **input** it uses.

### 03. Reviewer agreement on four pilot answers

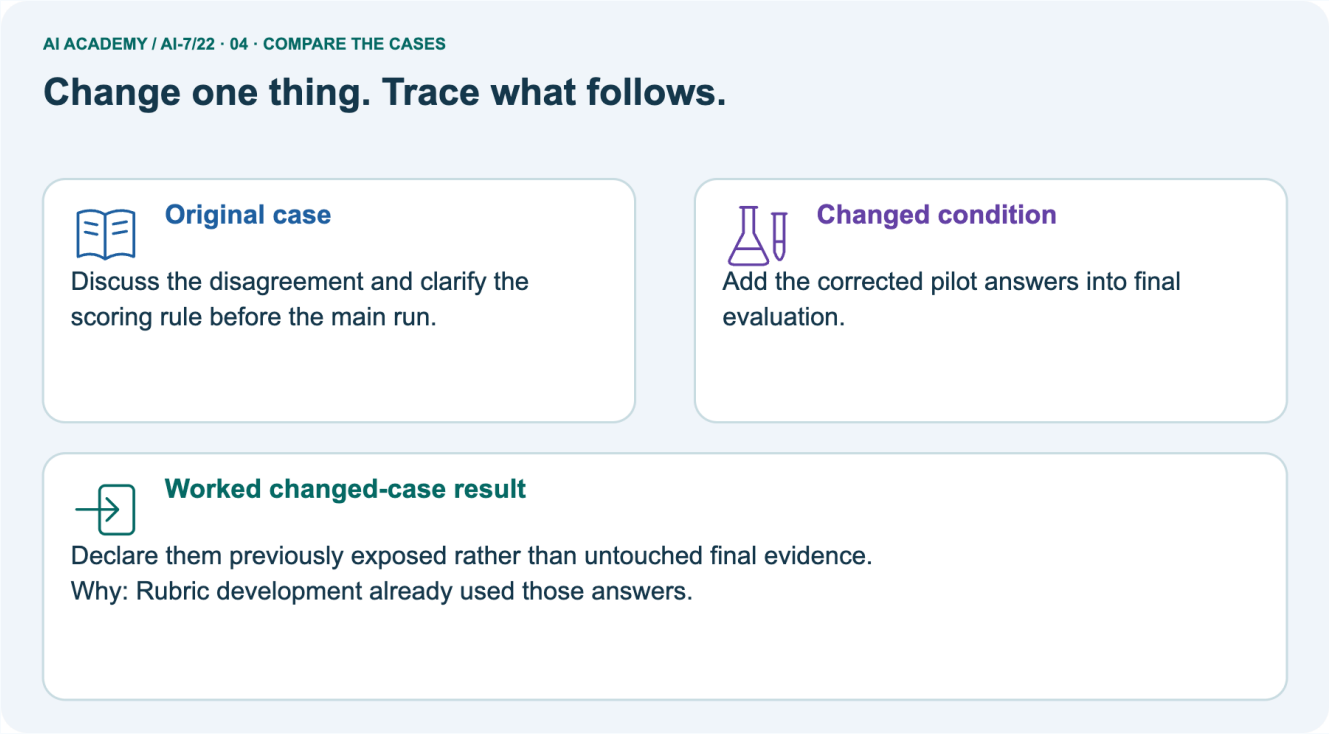



Follow the labelled picture with the matching explanation. Open the vector diagram

Reviewer agreement on four **pilot** answers. Units: **pilot** answers out of 4. Agree: 3. Disagree: 1.

> Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



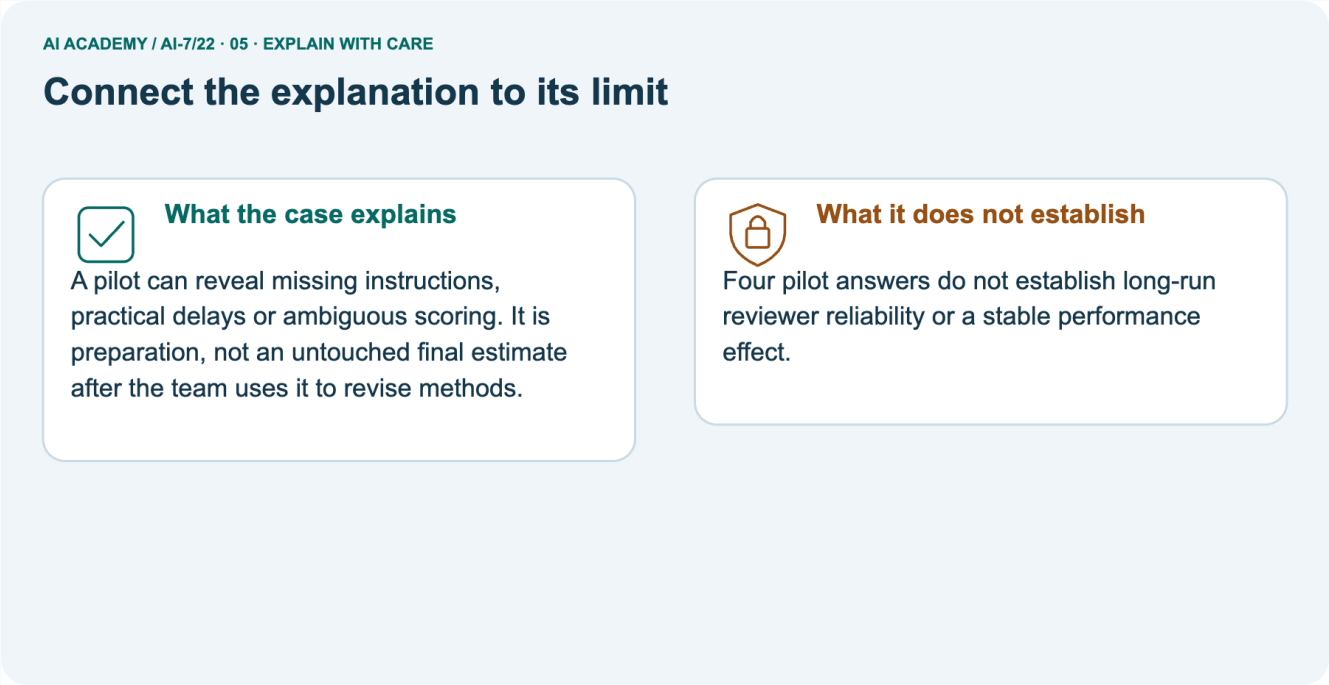

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Discuss the disagreement and clarify the scoring rule before the main run. Changed condition: Add the corrected **pilot** answers into final evaluation.

Expected result: Declare them previously exposed rather than untouched final **evidence**. Reason: Rubric development already used those answers.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit



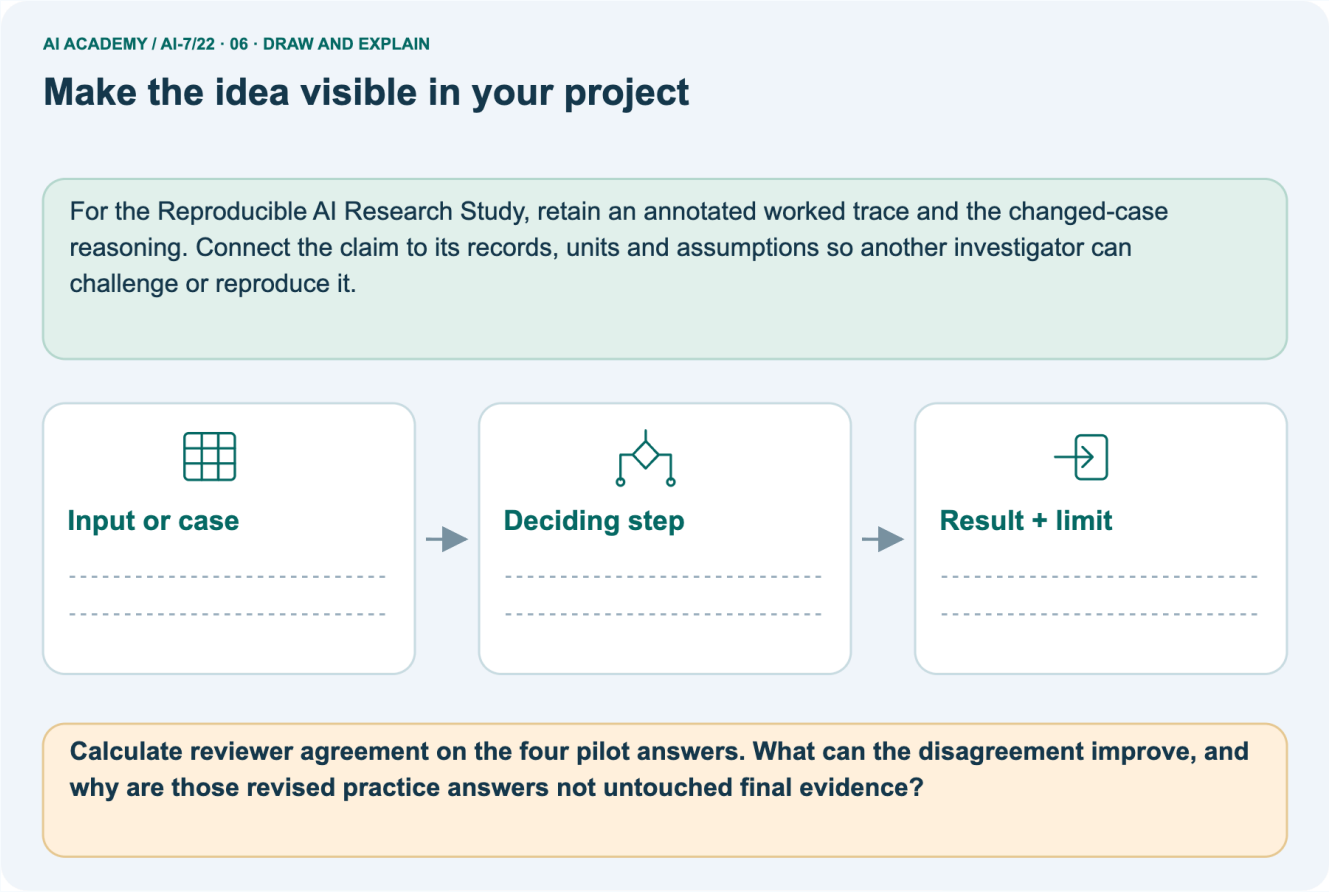

Follow the labelled picture with the matching explanation. Open the vector diagram

A **pilot** can reveal missing instructions, practical delays or ambiguous scoring. It is preparation, not an untouched final estimate after the team uses it to revise methods.

Limit: Four **pilot** answers do not establish long-run reviewer reliability or a stable performance effect.

> Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate reviewer agreement on the four **pilot** answers.

What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.